

UNIVERSITY OF EDINBURGH  
COLLEGE OF SCIENCE AND ENGINEERING  
SCHOOL OF INFORMATICS

**INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS**

**Wednesday 13<sup>th</sup> May 2015**

**09:30 to 11:30**

**INSTRUCTIONS TO CANDIDATES**

**Answer any TWO questions.**

**All questions carry equal weight.**

**CALCULATORS MAY BE USED IN THIS EXAMINATION**

Year 3 Courses

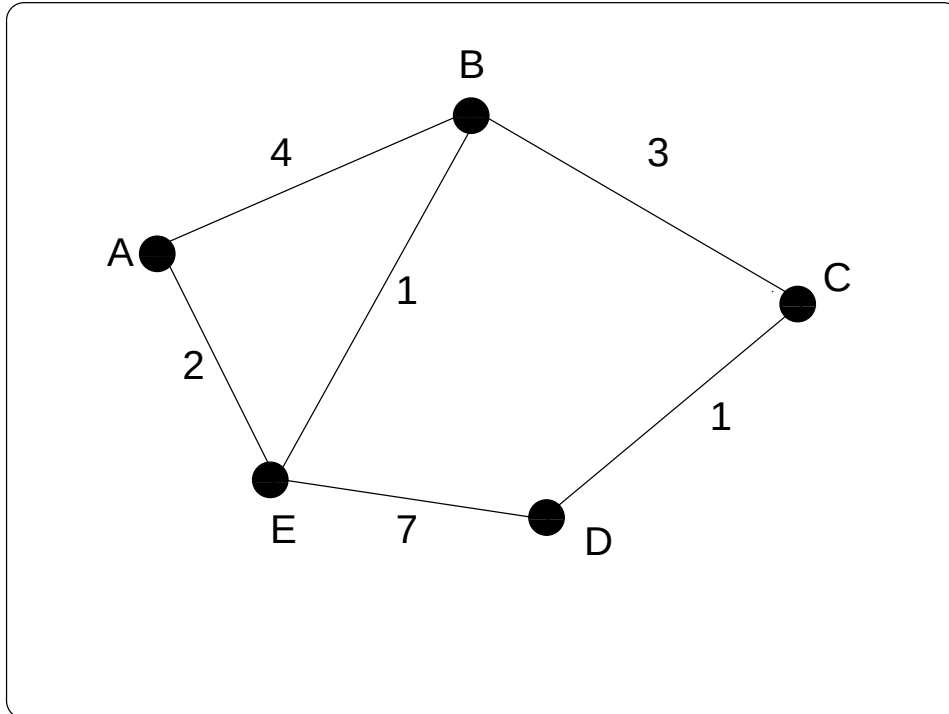
Convener: S. Viglas

External Examiners: A. Cohn, T. Field

**THIS EXAMINATION WILL BE MARKED ANONYMOUSLY**

1. (a) What are the advantages and disadvantages of protocol layering? [5 marks]
- (b) Discuss the differences between connectionless (UDP) demultiplexing and connection-oriented (TCP) demultiplexing. [4 marks]
- (c) Under what circumstances does non-persistent HTTP outperform persistent HTTP in terms of web page download time? [3 marks]
- (d) Consider a home Internet access scenario. To speed up web page download times and make efficient use of your Internet access connection, you decide to enable caching within your web browser. How does this compare with the alternative of web caching proxy server installed within the infrastructure of your Internet Service Provider (ISP)? [4 marks]
- (e) Suppose you acquired a new domain for your organisation called mynew-domain.org. How would the DNS system be made aware of it so that any requests to access www.mynewdomain.org are received by the web server of your domain? [3 marks]
- (f) Consider the effect of using slow start on a link with a 20ms round-trip time and no congestion. The receive window is 16KB and the maximum segment size is 1460 bytes. How long does it take before the first full window can be sent in one transmission round? [2 marks]
- (g) Consider sending a large file from one host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start. Assume *CongWin* increases by 1 MSS (maximum segment size) every time a batch of ACKs is received. Also assume approximately constant RTTs (round-trip times).
  - i. How long does it take for *CongWin* to increase from 1 MSS to 5 MSS assuming no loss events? [2 marks]
  - ii. What is the average throughput in terms of MSS and RTT for this connection when the *CongWin* is incremented from 1MSS to 5MSS, as per part i. above? [2 marks]

2. (a) What is the difference between routing and forwarding? [2 marks]
- (b) List five key differences between IPv4 and IPv6. [5 marks]
- (c) What is network address translation (NAT)? [2 marks]
- (d) Outline the key considerations that drive the inter autonomous system (AS) and intra-AS routing in the Internet. [2 marks]
- (e) Suppose a router has the following CIDR entries in its routing table:
- | Net/Prefix     | Next Hop    |
|----------------|-------------|
| 135.46.56.0/22 | Interface 0 |
| 135.46.60.0/22 | Interface 1 |
| 192.53.40.0/23 | Router 1    |
| default        | Router 2    |
- For each of the following IP addresses, what does the router do if a packet with that address arrives? (i) 135.46.63.10; (ii) 135.46.57.14; (iii) 135.46.52.2; (iv) 192.53.40.7; (v) 192.53.56.7. [5 marks]
- (f) Describe a general mechanism for resolving the counting-to-infinity problem. [5 marks]
- (g) The figure shown below indicates the state of the network topology and link costs at node A at the end of flooding-based link state dissemination (i.e., using the link-state broadcast algorithm). Determine the shortest path from node A to every other node using Dijkstra's shortest path algorithm. Show all steps of the algorithm.



[4 marks]

3. (a) Compare cable and ADSL based broadband access networks. [3 marks]
- (b) As each network interface at a router or host is assigned a unique IP address, why is a separate MAC address (also called a physical address) needed? [2 marks]
- (c) What is the difference between link bandwidth and link throughput? Under what conditions, do they become almost identical? [4 marks]
- (d) Compare and contrast the TCP reliable data transfer approach with the go-back-N and selective repeat protocols. [4 marks]
- (e) An upper-layer packet is split into 5 frames, each of which has an 80 percent chance of arriving undamaged. If no error control is performed by the data link protocol, how many times must the message be sent on average to get the entire message through? [4 marks]
- (f) Why is slotted ALOHA two times more efficient than its unslotted variant? [4 marks]
- (g) Switches have the nice property that they learn their forwarding tables via data packets without the need for a routing protocol. Can routers follow the same approach to construct and maintain their forwarding tables? Discuss your answer. [4 marks]